

Open Phase Detection (OPD) Guide

Power Application Controller™





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1 INTRODUCTION

This document describes the feature of open phase detection (OPD) for both open loop (OL) and close loop (CL) operation. The detection algorithms for OPD OL and OPD CL are independent, which means each feature can be enabled individually according to the customer's needs. The OPD OL can be also called as OPD during startup. The detection threshold can be configured to adjust the detection sensitivity.

2 OPD OVERVIEW

OPD OL will be explained followed by the OPD CL. Both OPDs have tuning parameters to be configured. The flowcharts of both OPDs are described, respectively.

3 OPD DURING OPEN LOOP

3.1 Tuning parameters

To enable OPD OL, the following `#define` in "config_features.h" needs to be enabled:

```
#define ENABLE_OPD_STARTUP
```

The time delay to sample the current during OPD can be configured by the below `#define`:

```
#define OLOPD_STARTSAMPLEDELAY    500
```

The actual delay time is calculated as (switching_cycle * OLOPD_STARTSAMPLEDELAY).

It is worthy to point out that the total sample as below is not recommended to change because the OPD lib would need to be recompiled.

```
#define OLOPD_TOTALSAMPLES        1024
```

3.2 OPD OL flowchart

There are two state machines for OPD OL. One is "statemachine_opdol_main()" (running at systick), which is responsible for determining the OPD based on the sampled current. The other is "statemachine_opdol_ctl()" (running at control cycle), which is used to sample and store each phase current. The flowchart of OPD OL is illustrated in Figure 3.2.

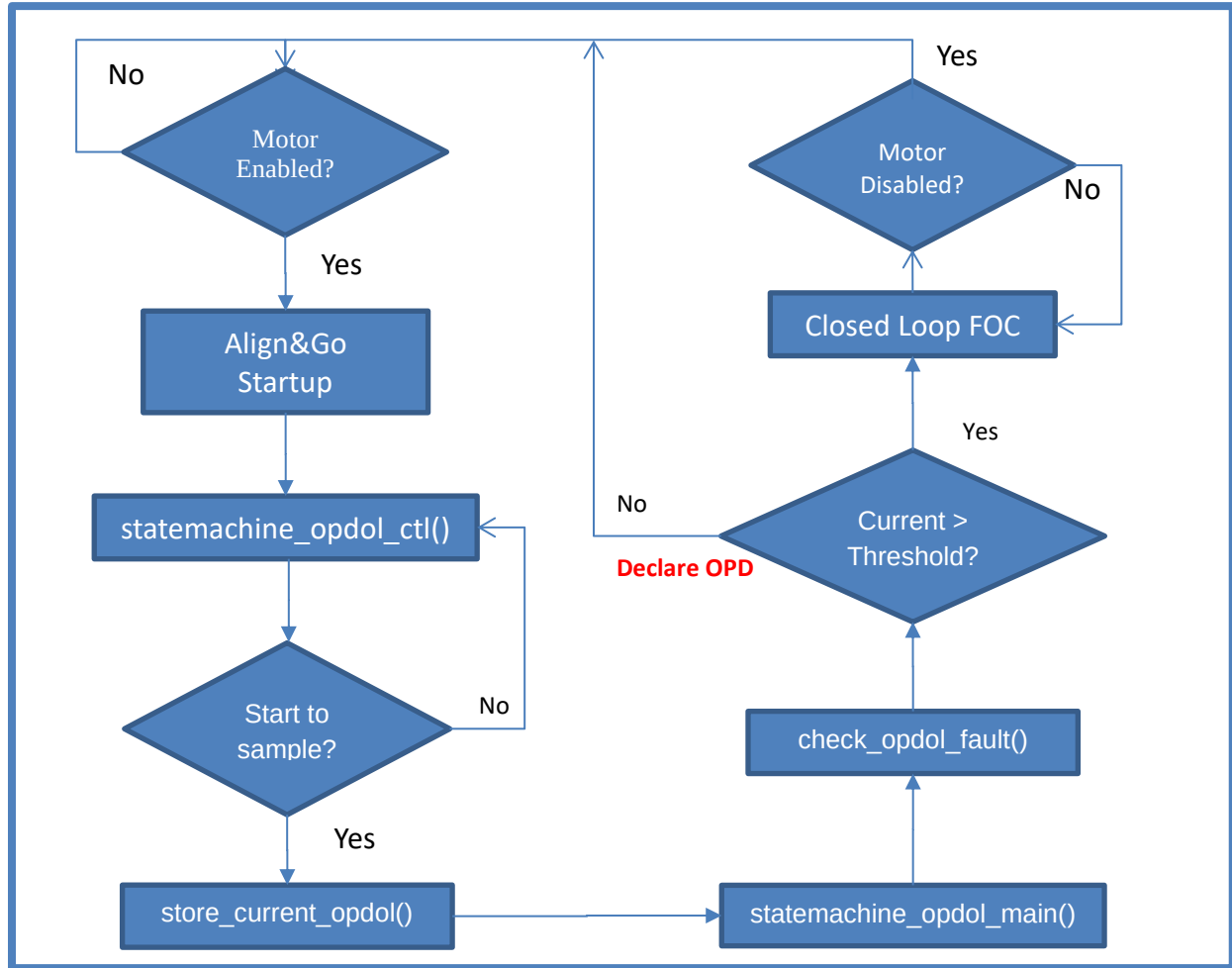


Figure 3.2 Flowchart of OPD OL

4 OPD DURING CLOSE LOOP

4.1 Tuning parameters

To Enable OPD CL, the following #define in “config_features.h” needs to be enabled:

```
#define ENABLE_OPD_CL
```

The minimum speed to start checking OPD can be configured by the following #define:

```
#define OPDCL_MINSPEED_RADPS_Q16 QCONST((1.0f * FLT_2PI), Q16)
```

Furthermore, the following #define can be used to configure OPD sensitivity:

```
#define OPDCL_COUNT_THRESHOLD 5 // This is for the majority filter
#define OPDCL_THRESHOLD_PCT Q3_0P750 // Phase imbalance percentage
#define OPDCL_CURRENT_THRESHOLD_Q14 QCONST(0.05f, Q14)
```

// Current threshold for declaring OPD

4.2 OPD CL flowchart

As similar as OPD OL, there are two main for OPD CL too. One is “check_opdcl_fault ()” (running at systick), which is responsible for determining the OPD based on the sampled current. The other is “integrate_and_log_phase_current()” (running at control cycle), which is used to sample and store each phase current. The flowchart of OPD CL is illustrated in Figure 4.2.

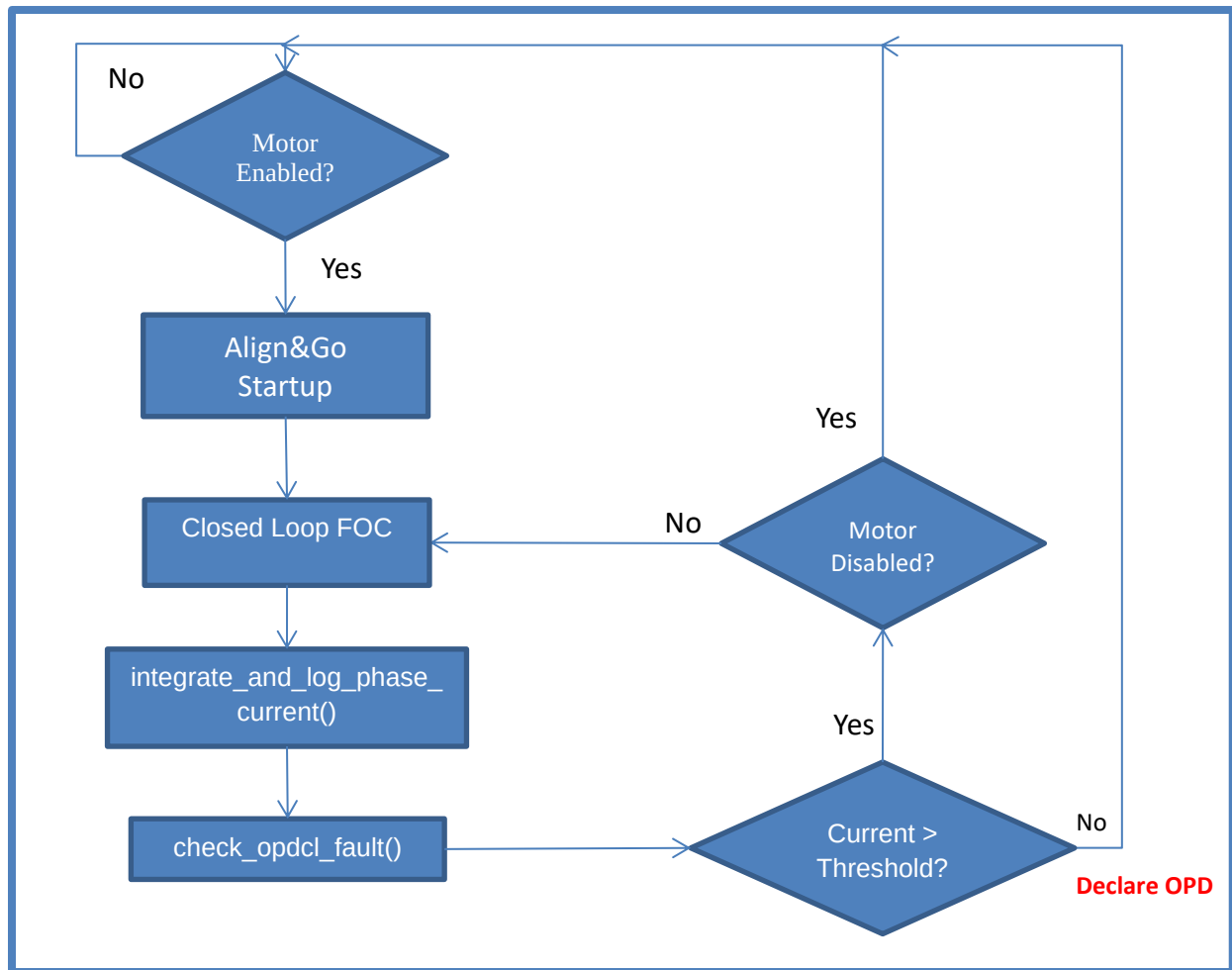


Figure 4.2 Flowchart of OPD CL



CONTACT INFORMATION

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